CS 5326: Advanced Generative AI and Agents

Programming Assignment 2

Instruction Tuning and Direct Preference Optimization on a Small Language Model

Final Report

Instructor	Dr Agha Ali Raza
Student	Syed Muhammad Ashhar Shah
Roll number	25030009
Program	MSCS

1 Repository, Models, and Data

The GitHub repository `https://github.com/AshharShah/small-model-alignment` holds all my code, configurations, metrics, generated responses, pairwise comparison files, and judged outputs. The limitations and reproducibility section lists the key paths.

Models. The base checkpoint is Qwen/Qwen2.5-0.5B [1]. I fine-tuned it for SFT, then trained both DPO models from the best SFT checkpoint, which also serves as the frozen reference model in the DPO loss [2]. For the LoRA persona task, the assignment requires a different model, so I used the instruction-tuned HuggingFaceTB/SmolLM2-360M-Instruct [3].

Training and evaluation data. Each training track has its own data (Table 1). SFT uses 30,000 training and 1,000 validation examples, each DPO run uses 15,000 training and 1,000 validation preference pairs, and the LoRA task uses 198 training and 22 validation pairs. Table 2 lists the four fixed evaluation sets: the course provides the HH set, and I built the other three. Within each comparison, I kept the evaluation prompts and their order fixed, so every model saw exactly the same inputs.

Table 1. Training data per track. I shuffled every dataset with seed 42 before filtering.

Track	Source (train / validation split)	Train / val	Filtering
SFT	UltraChat-200k [4] (`train_sft` / `test_sft`)	30,000 / 1,000	first user–assistant turn only; prompt + response + EOS $\leq$ 512 tokens (this limit drops 21,560 of 51,565 train rows)
HH-RLHF DPO	HH-RLHF [5] (`train` / `test`)	15,000 / 1,000	single-turn only (drops 34,382 multi-turn rows); prompt $\leq$ 256 and prompt + response + EOS $\leq$ 512 tokens (drops 127 more)
UltraFeedback DPO	UltraFeedback, binarized [6, 7] (`train_prefs` / `test_prefs`)	15,000 / 1,000[†]	single-turn only; chosen score $>$ rejected (drops 3,687 ties); same length limits (drops 12,124 more)
LoRA persona DPO	provided `ramsay_style_preferences.json`	198 / 22	none; a seeded shuffle (seed 42) holds out 22 rows

[†] 27 of the 1,000 validation pairs come from train rows that I did not use for training, because the test split has too few usable single-turn pairs.

Table 2. Fixed evaluation sets. "Alignment tax" means evaluating a DPO model on the other track's prompts.

Evaluation set	Prompts	Composition	Used for
AlpacaEval-300	300	random subset (seed 42) of the 805 AlpacaEval prompts [8]: self-instruct 95, Koala 60, OASST 54, helpful_base 51, Vicuna 40	SFT vs. DPO-UF (main), DPO-HH (alignment tax)
HH-eval-300	300	synthetic helpful/harmless set from the assignment; 10 categories $\times$ 30 prompts	SFT vs. DPO-HH (main), DPO-UF (alignment tax)
Persona-40	40	hand-written; 4 categories $\times$ 10 prompts (cooking, coding, general help, safety-sensitive); no overlap with the training prompts	LoRA adapters vs. base model (test)
Held-out-22	22	validation rows of the persona data that no training run used	LoRA adapter selection (dev)

Seeds, filtering and prompt format. I used seed 42 everywhere to keep the experiments reproducible: for dataset shuffling and subset selection (including the AlpacaEval subset and the persona train/validation split), for every training run, and for the A/B order in the pairwise comparison files.

All Qwen runs share one prompt format, `### Instruction:` ⟨instruction⟩ `### Response:`, with each marker on its own line and no chat template. This format covers SFT, both DPO runs, and every generation. SmolLM2 uses its own chat template for both training and generation.

I appended an EOS token to every training response and kept only the examples whose prompt, response, and EOS fit within 512 tokens. As a result, no example needed truncation and every response keeps its EOS

label. I also kept only single-turn data: the first exchange of each UltraChat conversation, single-turn pairs from HH-RLHF and UltraFeedback, and DPO prompts of at most 256 tokens.

2 Experimental Configuration

Table 3 reports the values I used for the four submitted checkpoints, not the guide defaults. I generated every number in it from the saved configuration file of the matching run, so the table describes the submitted checkpoints exactly. I trained on Kaggle T4 GPUs, with the DPO policy on one GPU and its frozen reference on the other.

Table 3. Configuration of the four submitted checkpoints. "Kept" marks the checkpoint I submitted.

Run	Trainable parameters	Optimizer and schedule	Batch, precision, length, seed	DPO β and reference	Generation
SFT	full fine-tuning (all weights)	AdamW, $\beta = (0.9, 0.95)$ peak LR 2×10^{-5} cosine to 0, warmup 28 steps (3%) weight decay 0.1, clip 1.0	2×16 accum. $= 32$ fp32 weights, fp16 autocast max length 512 1 epoch, 937 steps (kept: step 937) seed 42	n/a	greedy decoding 512 new tokens repetition penalty 1.0 length penalty 1.0 plain prompt template
HH-RLHF DPO	full fine-tuning (all weights)	AdamW, $\beta = (0.9, 0.95)$ peak LR 1×10^{-6} constant, no warmup weight decay 0, clip 1.0	1 pair $\times$ 32 accum. $= 32$ policy fp32, reference fp32, both under fp16 autocast max length 512 1 epoch, 468 steps (kept: step 400) seed 42	$\beta = 0.1$ reference: frozen SFT best checkpoint (step 937)	greedy decoding 512 new tokens repetition penalty 1.0 length penalty 1.0 plain prompt template
UltraFeedback DPO	full fine-tuning (all weights)	AdamW, $\beta = (0.9, 0.95)$ peak LR 1×10^{-6} constant, no warmup weight decay 0, clip 1.0	1 pair $\times$ 32 accum. $= 32$ policy fp32, reference fp32, both under fp16 autocast max length 512 1 epoch, 468 steps (kept: step 468) seed 42	$\beta = 0.1$ reference: frozen SFT best checkpoint (step 937)	greedy decoding 512 new tokens repetition penalty 1.0 length penalty 1.0 plain prompt template
LoRA-DPO (TRL)	LoRA $r = 16$, $\alpha = 32$, dropout 0.05 targets: q, k, v, o, gate, up, down 8,683,520 of 370,504,640 trainable (2.34%)	AdamW, $\beta = (0.9, 0.999)$ (Trainer default) peak LR 5×10^{-5} cosine to 0, warmup 3 steps weight decay 0, clip 1.0	4×4 accum. $= 16$ fp32 weights, fp16 autocast max length 512 3 epochs, 39 steps seed 42	$\beta = 0.1$, plus NLL on the chosen answer ($\text{rpo}_\alpha = 1$) reference: base SmolLM2, adapter disabled	greedy decoding 512 new tokens repetition penalty 1.0 length penalty 1.0 SmolLM2 chat template

SFT and both DPO runs share the same optimizer, gradient clipping, maximum length, and seed. The two DPO runs differ only in their data: they start from the same SFT checkpoint, which also serves as their frozen reference. I kept the checkpoint with the best validation reward accuracy, which was step 400 of 468 for HH-RLHF and the final step for UltraFeedback. The best SFT checkpoint is also the final one (step 937).

The LoRA run differs in more than the data. It trains a LoRA adapter on SmolLM2 with TRL's `DPOTrainer`, using a larger learning rate, a smaller batch, and an extra negative log-likelihood (NLL) term on the chosen answer. The next section explains why I added that term.

Generation. Every evaluation used greedy decoding (`do_sample=False`), so temperature and top-p do not apply. I generated up to 512 new tokens with repetition penalty 1.0, length penalty 1.0, and no n-gram blocking, and I used identical settings for SFT and DPO within each comparison so that the judge compares models, not decoding policies. The Qwen runs use the plain prompt template from Section 1, and the SmolLM2 runs use the model's chat template.

References

[1] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

[2] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *arXiv preprint arXiv:2305.18290*, 2023.

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[5] Yuntao Bai, Andy Jones, Kamal Ndousse, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022.

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[8] Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. Length-controlled AlpacaEval: A simple way to debias automatic evaluators. *arXiv preprint arXiv:2404.04475*, 2024.